

Working Life Balance Dashboard – Report

1. Introduction

This report explains the **Working Life Balance Dashboard**, which analyzes how daily lifestyle factors such as **work, sleep, exercise, and rest** influence **lifespan (age at death)**.

The dashboard is built using:

- **Python** – data cleaning and preprocessing
- **SQL** – data analysis
- **Power BI** – interactive visualizations

The dataset contains **10,000 individuals** and represents realistic daily time allocation close to **24 hours**, making it suitable for lifestyle and longevity analysis.

2. Key Performance Indicators (KPIs)

At the top of the dashboard, KPI cards provide a quick summary:

- **Avg Lifespan (Years)**: Shows the overall average age at death
- **Avg Work Hours / Day**: Average daily working time
- **Avg Sleep Hours / Day**: Average daily sleep duration
- **Avg Exercise Hours / Day**: Average time spent on physical activity
- **Total Individuals**: Total number of records analyzed

📌 Purpose:

These KPIs give a high-level overview before diving into detailed analysis.

3. Impact of Work Hours on Lifespan (Scatter Plot)

Chart: Scatter Plot

Relation: Work Hours → Age at Death

This chart shows how **daily work hours affect lifespan**.
Each dot represents an individual, colored by **occupation type**.

Insight:

- Individuals working **very long hours** tend to have a **lower lifespan**

- People with **moderate work hours** generally live longer
- The relationship is **non-linear**, indicating diminishing health returns from overworking

📌 Tooltips display **sleep and exercise hours**, helping explain why individuals with similar work hours may have different lifespans.

4. Average Lifespan by Occupation (Bar Chart)

Chart: Bar Chart

Relation: Occupation → Age at Death

This chart compares the **average lifespan across different occupations**.

Insight:

- Certain professions show **better longevity**, likely due to balanced routines
- High-stress or physically demanding occupations may show lower average lifespan

📌 Helps identify how **professional lifestyle impacts health outcomes**.

5. Sleep Duration vs Lifespan (Line Chart)

Chart: Line Chart

Relation: Sleep Hours → Age at Death

This visualization highlights the effect of **sleep duration on longevity**.

Insight:

- **Moderate sleep (around 7-8 hours)** is associated with higher lifespan
 - Too little or excessive sleep shows reduced benefits
 - Demonstrates **diminishing returns** in health outcomes
-

6. Exercise vs Lifespan (Column Chart)

Chart: Column Chart

Relation: Exercise Hours → Age at Death

This chart shows how **physical activity influences lifespan**.

Insight:

- Regular, moderate exercise correlates with **higher longevity**
- Extremely low or excessive exercise provides fewer benefits

✦ Reinforces the importance of a **balanced lifestyle**.

7. Gender Distribution (Donut Chart)

Chart: Donut Chart

Relation: Gender → Population Count

This chart shows the **gender composition** of the dataset.

Insight:

- Confirms balanced representation
 - Provides context for gender-based comparisons
-

8. Average Lifespan by Gender (Bar Chart)

Chart: Bar Chart

Relation: Gender → Age at Death

This chart compares **average lifespan between genders**.

Insight:

- Highlights observable differences in longevity
 - Useful for demographic-level analysis
-

9. Daily Time Allocation (Stacked Column Chart)

Chart: Stacked Column Chart

Relation: Daily Time Allocation → Occupation

This chart shows how different occupations divide their **24-hour day** into:

- Work
- Sleep
- Rest
- Exercise

Insight:

- Clear **work-life balance differences** across professions
 - Occupations with balanced time distribution show healthier patterns
-

10. Interactive Filters (Slicers)

The dashboard includes interactive filters for:

- **Occupation Type**
- **Gender**
- **Age at Death (Range)**

Benefit:

- All charts update dynamically
 - Users can explore specific groups and patterns
 - Enhances decision-making and analysis
-

11. Conclusion

The **Working Life Balance Dashboard** demonstrates that:

- Balanced work hours improve longevity
- Sleep and exercise play a critical supporting role
- Occupation significantly affects lifestyle patterns
- Health outcomes depend on **overall daily balance**, not a single factor

This dashboard helps users **visually understand lifestyle-longevity relationships** and supports **data-driven insights** for healthier work-life decisions.

“This dashboard shows how work, sleep, exercise, and rest together influence lifespan. By using interactive filters, we can easily identify which lifestyle patterns lead to better longevity.”